Nabila Alia Zahra	**Rangkuman Minggu 5** **Pengaplikasian ATMega 8535 ke Proteus**	Selasa, 27 September 2022
		16TNK5043 (Aplikasi Mikrokontroler)
3NK		Ferry Satria, BSEE.,M.T. Rahmawati Hasanah,S.ST.,M.T.

Mengenable semua pake SEI

Mengdisable semua pake CLI

Klo mau membuat

int 1 pake bit 7

Int 0 bit 6

Int 2 bit 5

Enable 0 pake logic 1 (GICR)

Misal int 1 mau di enable jadi 10000000

LDI R20, 0X80

Out gicr,r20

Klo mau int1 jadi rising edge (MCUCR)

LDI R20, 0X0C

OUT MCUCR,R20

Table 35. Interrupt 1 Sense Control

ISC11	ISC10	Description
0	0	The low level of INT1 generates an interrupt request.
0	1	Any logical change on INT1 generates an interrupt request.
1	0	The falling edge of INT1 generates an interrupt request.
1	1	The rising edge of INT1 generates an interrupt request.

Klo mau int 0 jadi low level

LDI R20,0

OUT MCUCR,R20

Table 36. Interrupt 0 Sense Control

ISC01	ISC00	Description
0	0	The low level of INT0 generates an interrupt request.
0	1	Any logical change on INT0 generates an interrupt request.
1	0	The falling edge of INT0 generates an interrupt request.
1	1	The rising edge of INT0 generates an interrupt request.

INTERRUPT VEKTOR

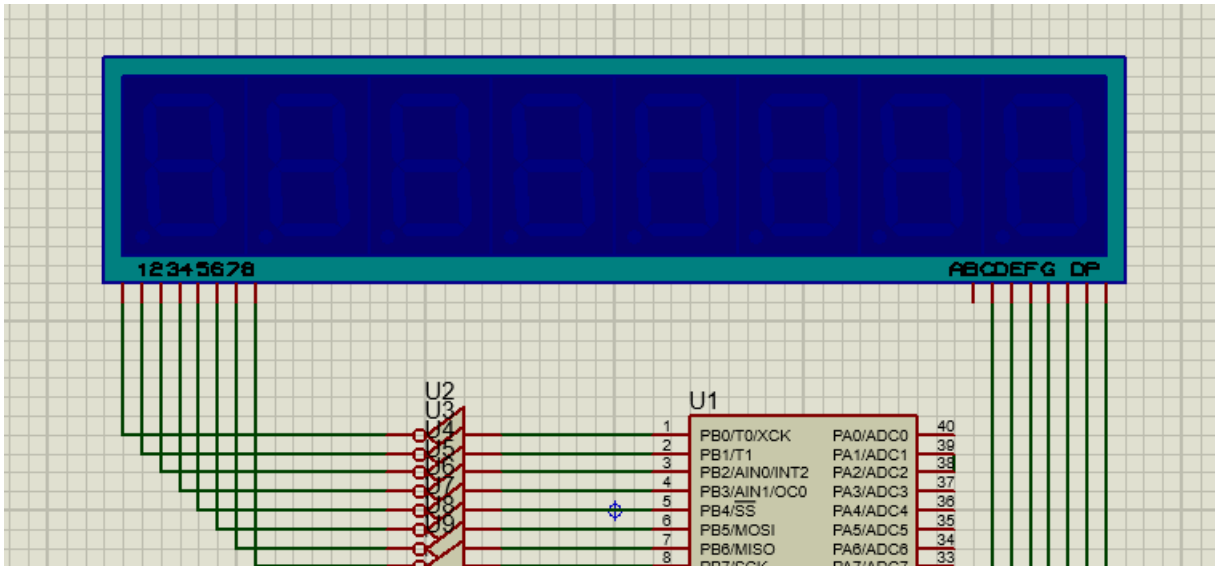
INT 0 : 001

INT 1 : 010

INT 2 : 011

Perintah untuk mengakhiri interrupt

Klo ada 8 7 segmen harus menghubungkan semua jadi Satu



12345678 :

ABCDEFG : common

Program menampilkan huruf Rajin

```
;Program Interrupt dan Timer
; pakai project "ROOT"
.include "m8535def.inc"
.org 0x00
.equ timer_1=0xFFF0
rjmp main

.org 0x001
rjmp int_0
```

```
main :
        ldi r16,low(ramend)
        out spl,r16
        ldi r16,high(ramend)
        out sph,r16

INT_1:
        cli
        ldi r17,0x20
        out gicr,r17  ; DISABLE SEMUA INTERRUPT

        ldi r17,0x01
        out mcucsr,r17  ;PILIH SISI NAIK ATAU SISI TURUN

        ldi r17,0x40  ; ENABLE GENERAL INT
        out gicr,r17

        ldi r17,0xFF   ; ATUR PORTB DAN PORTC SBG OUTPUT
        out ddrb,r17
        out ddrc,r17

        sei    ; SET GENERAL INT

        clr r20  ; R20 COUNTER

;====================
loop:
        ldi zl,low(2*posisi)
        ldi zh,high(2*posisi)

        ldi r26,$60  ; inisialisasi pointer x
        clr r27

lup:
        lpm r18,z+  ; Data Posisi dipindahkan ke Data Area
        st x+,r18
        cpi r26,$68
        brne lup

;======================
lup2:
        rcall nyala0
        andi r20,0x01 ; JIKA R20 GENAP, TDK BERKEDIP
        breq lup2          ; JIKA R20 GANJIL, BERKEDIP
        rcall mati0
        rjmp lup2

nyala0:
        cli   ; CLEAR INT
        ldi r29,$0F   ; R29=15

nyala:
```

```
        ldi r26,$60   ;POINTER X MENUNJUK KE DATA AREA
        clr r27      ; ------------------------------
        ldi zl,low(2*pola) ; POINTER Z menunjuk ke POLA KARAKTER
        ldi zh,high(2*pola)

nyala1:
        ld r28,x+
;       com r28
        out portb,r28

        lpm r21,z+
        out portc,r21

        rcall delay_2ms
        cpi r26,$68
        brne nyala1

        dec r29
        brne nyala
        sei
        ret
;=========================

;============================
mati0:
        cli
        ldi r29,125

mati:
        ldi r21,0xFF
        out portc,r21

        rcall delay_2ms

        dec r29
        brne mati
        sei
        ret
;=========================

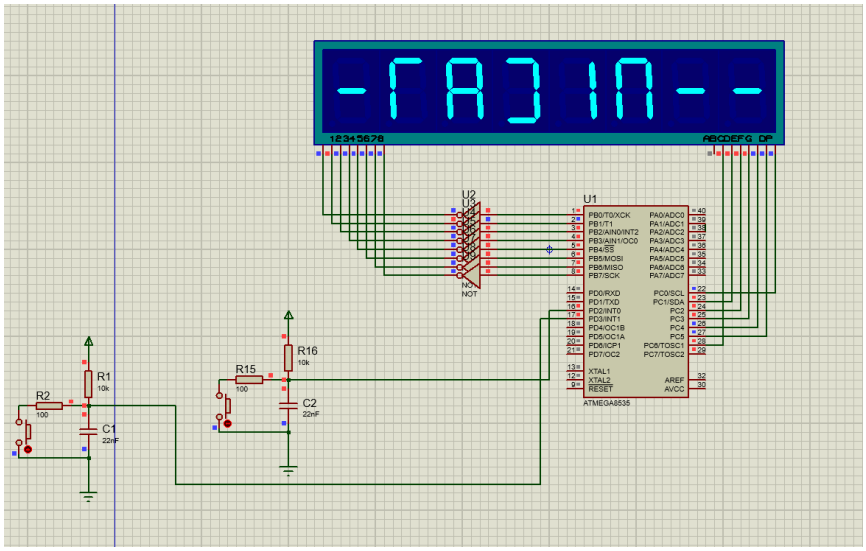
delay_2ms:
        cli
        push r16
        push r17
        ldi r16,4
        out timsk,r16
        ldi r16, high(timer_1)
        out tcnt1h,r16
        ldi r16, low(timer_1)
        out tcnt1l,r16
        ldi r16,5
        out tccr1b,r16
```

```
looptimer:
  in r17,tifr
  sbrs r17,tov1
  rjmp looptimer
  ldi r16,4
  out tifr,r16
  pop r17
  pop r16
  sei
  ret

INT_0:
        cli
        inc r20
        sei
        reti

pola:
.db 0xBF, 0xCE, 0x88, 0xF0, 0xF9, 0xC8, 0xBF, 0xBF;karakter apa yg akan dimunculkan

posisi:
.db 0xFE, 0xFD, 0xFB, 0xF7, 0xEF, 0xDF, 0xBF, 0x7F;digit mana yg digunakan, cmn bisa salah satu yg aktif gak bisa 8 8 nya, krn commonnya 1 per satu aktifnya (sistem scan?)
```



Program menampilkan huruf belajar

```
.INCLUDE"M8535DEF.INC"
.ORG 0
RJMP MAIN
                ; BCD UP-DOWN COUNTER MODULO 36, DENGAN 74LS47
                ; DELAY MENGGUNAKAN LOOP 3 REGISTER
        ; IMPLEMENTASI: PROTEUS DEC_COUNT_01 [DI PROTEUS_PROJECT]
```

```
MAIN:LDI R16,LOW(RAMEND)
        OUT SPL,R16
        LDI R16,HIGH(RAMEND)
        OUT SPH,R16

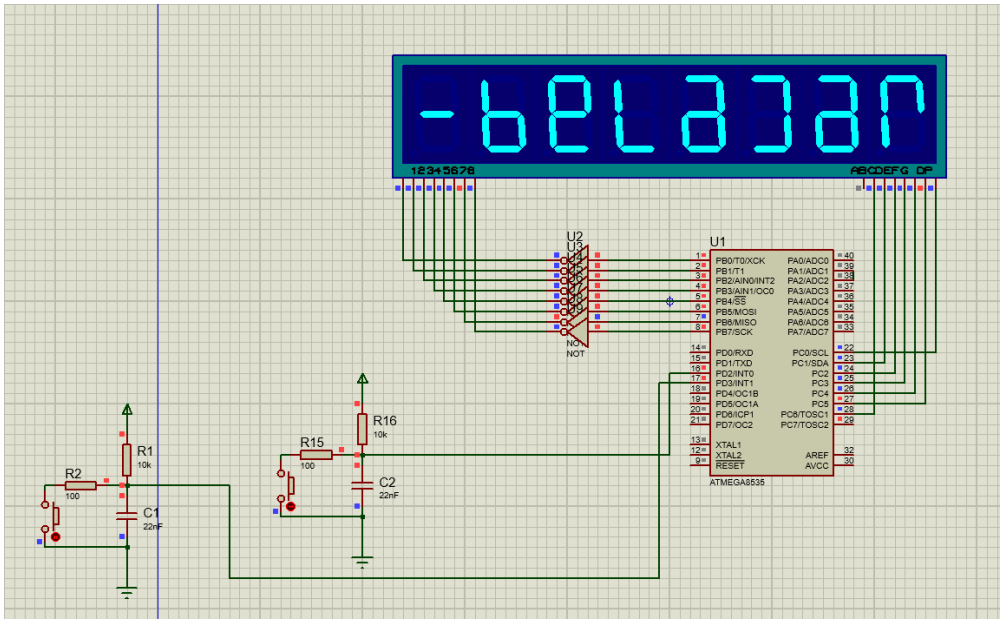
        LDI R20,0
        OUT DDRA,R20      ; PORT A SBG INPUT
        LDI R20,0XFF
        OUT DDRB,R20      ; PORT B SBG OUTPUT
        LDI R21,0X99      ; KOMPLEMEN 10 DARI 1
ITU:  LDI R18,0           ; INPUT AWAL UP COUNTER
SANA:   OUT PORTB,R18
        RCALL DELAY
        IN R17,PINA       ; MEMBACA DATA DARI PIN A
        MOV R16,R17
        ANDI R16,1
        CPI R16,0
        BRNE DOWN               ; JUMP KE DOWN COUNTER JIKA R16 = 1
UP:     LDI R20,1
        MOV R19,R18
        ADD R19,R20
        RCALL DAA
        MOV R18,R19
        CPI R18,0X36; BATAS ATAS UP COUNTER
        BRNE SANA
        RJMP ITU


INI:  LDI R18,0X35        ; NILAI AWAL DOWN COUNTER
DOWN:
SINI: OUT PORTB,R18
        RCALL DELAY
        IN R17,PINA       ; MEMBACA DATA DARI PIN A
        MOV R16,R17
        ANDI R16,1
        CPI R16,1
        BRNE UP                 ; JUMP KE UP COUNTER JIKA R16 = 0
        MOV R19,R18
        ADD R19,R21
        RCALL DAA
        MOV R18,R19
        CPI R18,0x99 ; BATAS BAWAH DOWN COUNTER
        BRNE SINI
        RJMP INI

DELAY:  LDI R23,3
LUP1: LDI R24,0XFF
LUP2: LDI R25,0
LUP3: DEC R25
        BRNE LUP3
        DEC R24
        BRNE LUP2
```

```
                DEC R23
                BRNE LUP1
                RET
daa:    push r16;amankan isi reg. ke stack
                push r17
                push r21
                push r24
                in r16,sreg; amankan sreg
                in r17,sreg
                ldi r24,0 ; r24 hanya sbg status dr carry flag
                adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
                out sreg,r17
                brhc lup4; lompat ke lup4 jika H=0
                rcall fk_ln; satuan + 06 jika H=1
        ;       rjmp lup5
lup4:   mov r21,r19
                andi r21,$0f; ambil dgt.satuan
                cpi r21,$0a ; > 9
                brcs lup5; lompat ke lup5 jika < 9
                rcall fk_ln; satuan + 06 jika > 9
lup5:   out sreg,r17; ambil, amankan kembali sreg, uji carry
                brcc lup6; lompat ke lup6 jika C=0
                rcall fk_hn; puluhan + 60 jika C=1
        ;       rjmp lup7
lup6:   mov r21,r19
                andi r21,0xf0; ambil dgt.puluhan
                cpi r21,0xa0
                brcs lup7; lompat ke lup7 jika pul < a
                rcall fk_hn; pul + 60 jika > 9
lup7:   out sreg,r16; mengatur carry flag sblm keluar:
                clc; C = 0
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
                out sreg,r16;ambil sreg, set c
                sec; C = 1
                in r16,sreg; amankan sreg stlh c=1
lup8:
                out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
                pop r17
                pop r16
                ret
fk_ln:  ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
                in r16,sreg
                brcc tuh
                ldi r21,0x01
                add r17,r21
```

```
tuh:    out sreg,r16
        adc r24,r24
        ret
fk_hn:  ldi r21,$60; koreksi dgt.puluhan
        out sreg,r16
        add r19,r21
        in r16,sreg
        adc r24,r24; amankan carry pd r24
        ret ; akhir dari prog. subrutin.
```



Buat program untuk menampilkan huruf A pada unit peraga 7 Segmen common Anoda, jika tombol int 0 ditekan akan menyebabkan : tampil a kecil selama 2 detik saja, pake push button untuk mengaktifkan int0.

```
.INCLUDE"M8535DEF.INC"
.ORG 0
RJMP MAIN
.ORG 1
RJMP INT_0

MAIN:LDI R16,LOW(RAMEND)
        OUT SPL,R16
        LDI R16,HIGH(RAMEND)
        OUT SPH,R16

        LDI R17,0XFF
        OUT DDRA,R17            ; PORT B SEBAGAI OUTPUT

        LDI R18,8               ; A KAPITAL PADA 7 SEGMENT CA
        OUT PORTA,R18

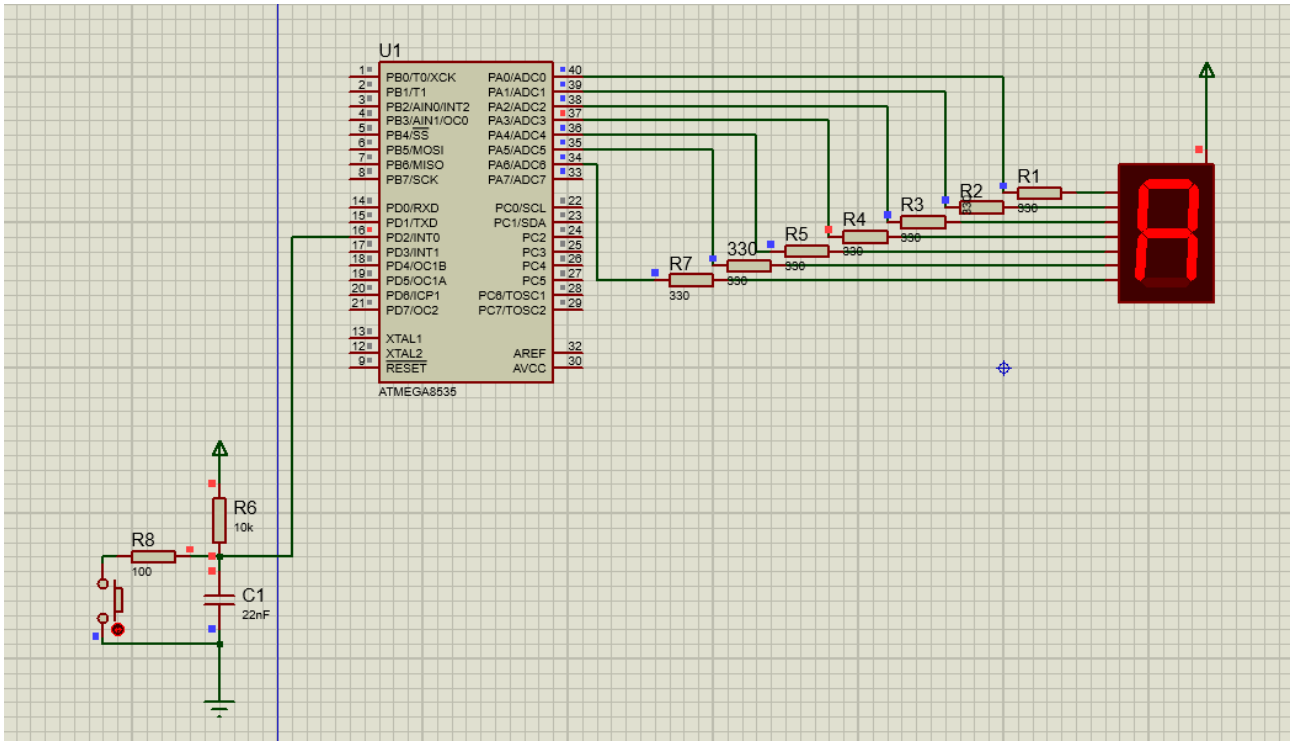
AKHIR: RJMP AKHIR
```

```
DELAY:      CLI
            LDI R23,3
LUP1: LDI R24,0XFF
LUP2: LDI R25,0
LUP3: DEC R25
            BRNE LUP3
            DEC R24
            BRNE LUP2
            DEC R23
            BRNE LUP1
            SEI
            RET


INT_0:CLI
            LDI R19,0X40
            OUT GICR,R19            ; GICR HANYA MENGENABLE INT0
            LDI R20,3
            OUT MCUCR,R20           ; INT0 AKTIF SAAT RISING EDGE

            LDI R26,0X20            ; A KECIL PADA 7 SEGMENT CA
            OUT PORTA,R26
            RCALL DELAY
            RCALL DELAY

            SEI
            RETI
```



Rancang suatu counter heksadesimal 1 digit modulo 16, kenaikan dikendalikan oleh tombol interrupt INT0, dengan seven segmen common katoda

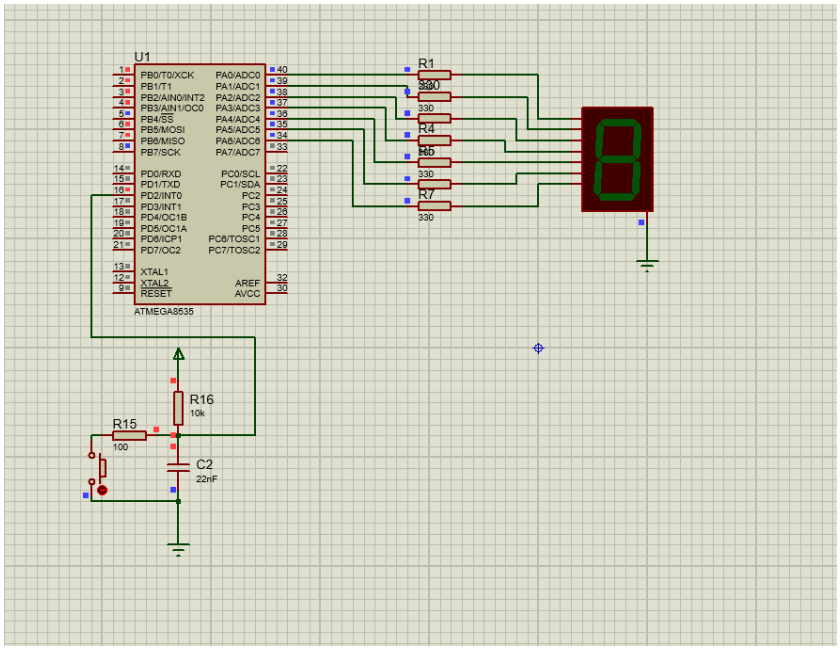
```
.include "m8535def.inc"
.org 0
RJMP MAIN

.ORG 0X01
RJMP INT_0
; Up-Counter Heksadesimal Modulo 16 (Menghitung secara berulang
; dari 0 sampai F); INT0 berfungsi sebagai Counter Incrementer
; Menggunakan Seven Segment Common Cathode.
; PROTEUS PROJECT: UP-CNT-INT0
;

MAIN:CLI
                LDI R20,LOW(RAMEND)
                OUT SPL,R20
                LDI R20,HIGH(RAMEND)
                OUT SPH,R20
                LDI R20,0XFF
                OUT DDRB,R20
                LDI R20,0X40
                OUT GICR,R20
                LDI R20,0X02
                OUT MCUCR,R20
                LDI R30,LOW(2*POLA) ; INIT POINTER Z
                LDI R31,HIGH(2*POLA)
                LDI R22,0 ; INIT COUNTER
                SEI
KK:             MOV R30,R22
                LPM R2,Z
                OUT PORTB,R2
                RCALL DELAY
                RJMP KK

DELAY:          CLI
                LDI R23,1
LP1:    LDI R24,4
LP2:    LDI R25,0
LP3:    DEC R25
                BRNE LP3
                DEC R24
                BRNE LP2
                DEC R23
                BRNE LP1
                SEI
                RET
                ;================
INT_0:;CLI
                INC R22
                CPI R22,0X10
```

```
                BRNE LK1
                LDI R22,0
LK1:;   SEI
                RETI
;==================
.ORG 0X80
POLA:
.DB 0X3F,0X06,0X5B,0X4F,0X66,0X6D,0X7D,0X07
.DB 0X7F,0X6F,0X77,0X7C,0X39,0X5E,0X79,0X71
```



Rancang suatu Up/Down Counter Heksadesimal 1 digit Modulo 16 (Menghitung naik/turun secara berulang dengan range output dari 0 sampai F). Gunakan seven segment jenis Common Cathode dengan tombol interupsi INT0 yang berfungsi untuk menaikkan/menurunkan nilai counter (Counter incrementer/decrementer); dan INT1 yang berfungsi sebagai pengatur Mode kerja Counter (up/down).

```
.include "m8535def.inc"
.org 0
RJMP MAIN

.ORG 0X01
RJMP INT_0

.ORG 0X02
RJMP INT_1
;Rancang suatu Up/Down Counter Heksadesimal 1 digit Modulo 16 (Menghitung naik/turun
;secara berulang dengan range output dari 0 sampai F). Gunakan seven segment jenis
;Common Cathode dengan tombol interupsi INT0 yang berfungsi untuk menaikkan/menurunkan
;nilai counter (Counter incrementer/decrementer); dan INT1 yang berfungsi  sebagai
;pengatur Mode kerja Counter (up/down).
```

```
MAIN:CLI
                LDI R16,LOW(RAMEND)
                OUT SPL,R16
                LDI R16,HIGH(RAMEND)
                OUT SPH,R16

                LDI R20,0XFF
                OUT DDRA,R20

                LDI R20,0XC0
                OUT GICR,R20

                LDI R20,0X0F
                OUT MCUCR,R20

                LDI R30,LOW(2*POLA) ; INIT POINTER Z
                LDI R31,HIGH(2*POLA)
                LDI R22,0 ; INIT COUNTER
                SEI

KK:             MOV R30,R22 ;MENAMPILKAN ISI COUNTER
                LPM R2,Z
                OUT PORTA,R2
                RCALL DELAY
                RJMP KK

DELAY:          CLI
                LDI R23,1
LP1:    LDI R24,4
LP2:    LDI R25,0
LP3:    DEC R25
                BRNE LP3
                DEC R24
                BRNE LP2
                DEC R23
                BRNE LP1
                SEI
                RET
                ;=================

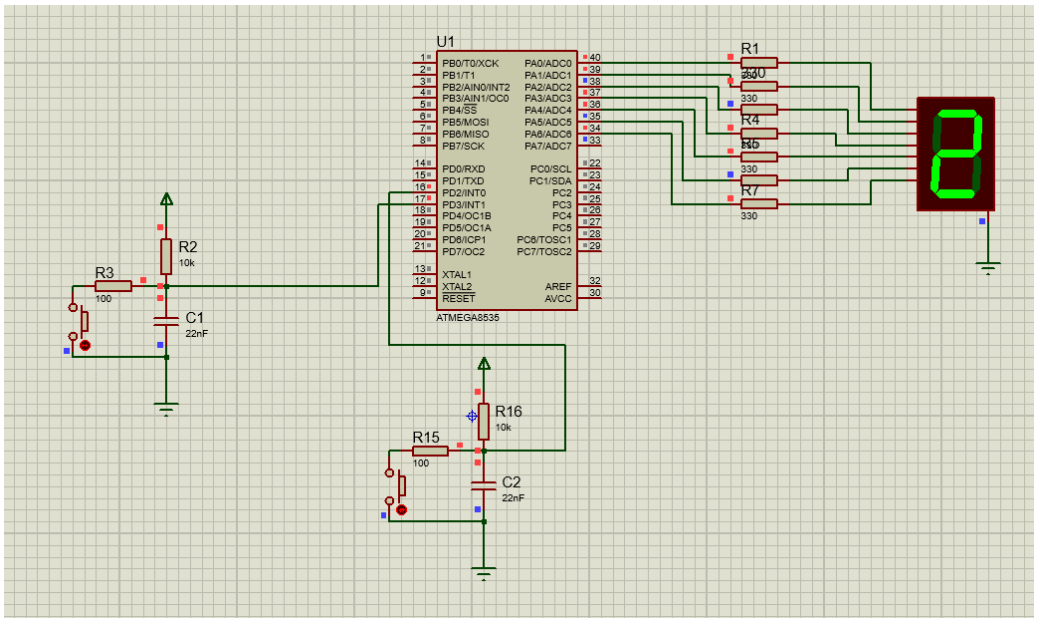
INT_0:CLI
                MOV R27,R20
                ANDI R27,1
                CPI R27,1
                BRNE TRN
                INC R22
                CPI R22,0X10
                BRNE LK1
                LDI R22,0
                RJMP LK1

TRN:    DEC R22
```

```
                CPI R22,0XFF
                BRNE LK1
                LDI R22,0X0F

LK1:;  SEI
                RETI

INT_1:CLI
                INC R20
                SEI
                RETI
;==================
.ORG 0X80
POLA:
.DB 0X3F,0X06,0X5B,0X4F,0X66,0X6D,0X7D,0X07
.DB 0X7F,0X6F,0X77,0X7C,0X39,0X5E,0X79,0X71
```



Rancang suatu Up Counter Heksadesimal 2 digit Modulo 45 (Menghitung naik secara berulang dari 00 sampai 2C). Gunakan seven segment jenis Common Anode dengan tombol interupsi INT2 yang berfungsi untuk menaikkan nilai counter (Counter incrementer).

```
.include "m8535def.inc"
.org 0
RJMP MAIN

.ORG 0X12
RJMP INT_2
; Up-Counter Heksadesimal Modulo 16 (Menghitung secara berulang
; dari 0 sampai F); INT0 berfungsi sebagai Counter Incrementer
; Menggunakan Seven Segment Common Cathode.
; PROTEUS PROJECT: UP-CNT-INT0
;
```

```
MAIN:CLI
                LDI R20,LOW(RAMEND)
                OUT SPL,R20
                LDI R20,HIGH(RAMEND)
                OUT SPH,R20

                LDI R20,0XFF
                OUT DDRA,R20

                LDI R20,0XFF
                OUT DDRC,R20

                LDI R20,0X20
                OUT GICR,R20

                LDI R20,0X40
                OUT MCUCSR,R20

                LDI R30,LOW(2*POLA) ; INIT POINTER Z
                LDI R31,HIGH(2*POLA)
                LDI R22,0 ; INIT COUNTER
                SEI

KK:             MOV R27,R22
                ANDI R27,0XF0
                LSR R27
                LSR R27
                LSR R27
                LSR R27

                MOV R30,R27
                LPM R2,Z
                OUT PORTA,R2

                MOV R27,R22
                ANDI R27,0X0F
                MOV R30,R27
                LPM R2,Z
                OUT PORTC,R2

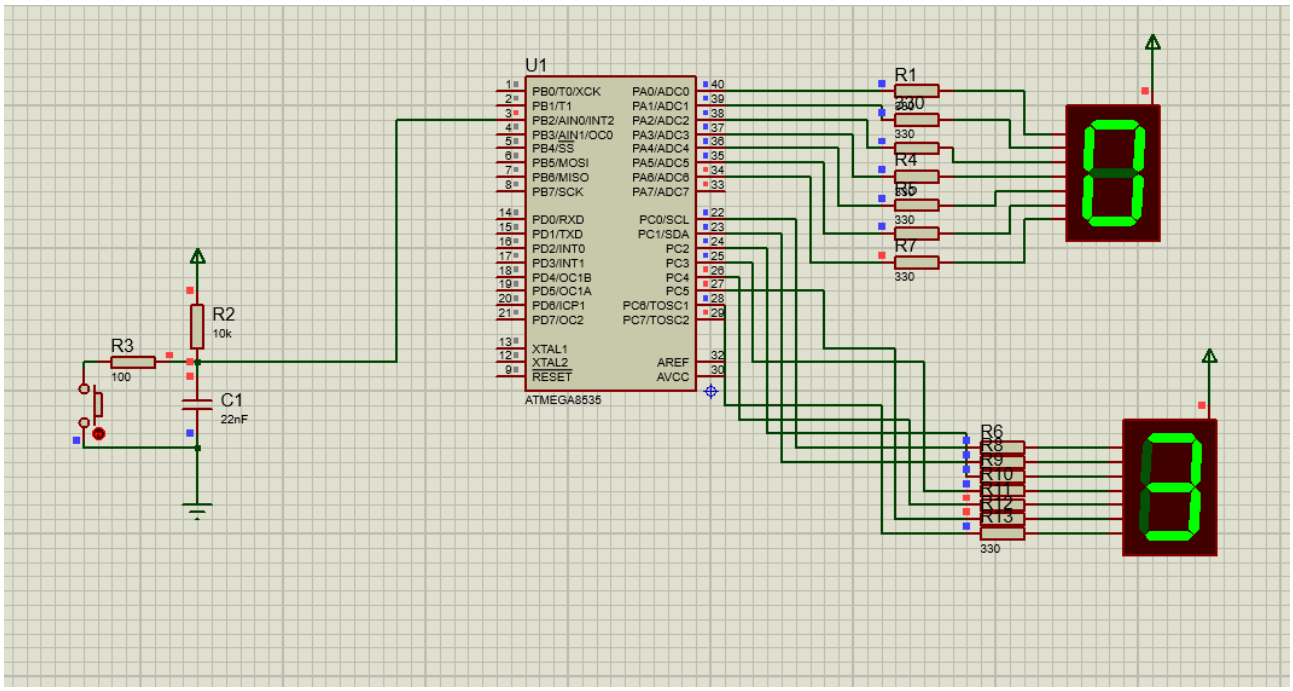
                RCALL DELAY
                RJMP KK

DELAY:          CLI
                LDI R23,1
LP1:    LDI R24,4
LP2:    LDI R25,0
LP3:    DEC R25
                BRNE LP3
                DEC R24
                BRNE LP2
                DEC R23
```

```
                BRNE LP1
                SEI
                RET
                ;=================
INT_2:CLI
                INC R22
                CPI R22,0X2D
                BRNE LK1
                LDI R22,0

LK1:    SEI
                RETI


;==================
.ORG 0X80
POLA:
.DB 0XC0,0XF9,0XA4,0XB0,0X99,0X92,0X82,0XF8
.DB 0X80,0X90,0X88,0X83,0XC6,0XA1,0X86,0X8E
```



Rancang sebuah program dengan INT 0 sebagai untuk menaikan dan menurunkan nilai counter, INT 2 sebagai mode kerja counter, dan INT 1 mengubah modulus counter moder 21 dan mode 35

```
.include "m8535def.inc"
.org 0
RJMP MAIN

.ORG 0X001
RJMP INT_0

.ORG 0X002
```

```
RJMP INT_1

.ORG 0X012
RJMP INT_2
; Up/Down-Counter Heksadesimal Modulo 21 dan 35
; INT0 berfungsi sebagai Counter Incrementer
; Menggunakan Seven Segment Common Anode INT1 mengubah modulo. INT2 mode kerja counter
; PROTEUS PROJECT: UP-CNT-INT0


MAIN:CLI
        LDI R20,LOW(RAMEND)
        OUT SPL,R20
        LDI R20,HIGH(RAMEND)
        OUT SPH,R20
        LDI R20,0XFF
        OUT DDRA,R20
        OUT DDRC,R20
        LDI R20,0XE0
        OUT GICR,R20
        LDI R20,0X0F
        OUT MCUCR,R20
        LDI R20,0X40
        OUT MCUCSR,R20
        LDI R30,LOW(2*POLA) ; INIT POINTER Z
        LDI R31,HIGH(2*POLA)
        LDI R22,0               ; INIT COUNTER
        SEI
KK:     MOV R27,R22
        ANDI R27,0XF0           ; MENGAMBIL ENAMBELASAN
        LSR R27
        LSR R27
        LSR R27
        LSR R27
        MOV R30,R27
        LPM R2,Z
        OUT PORTA,R2
        MOV R27,R22
        ANDI R27,0X0F           ; MENGAMBIL SATUAN
        MOV R30,R27
        LPM R2,Z
        OUT PORTC,R2
        RCALL DELAY
        RJMP KK


DELAY:  CLI
        LDI R23,1
LP1:  LDI R24,4
LP2:  LDI R25,0
LP3:  DEC R25
```

```
                BRNE LP3
                DEC R24
                BRNE LP2
                DEC R23
                BRNE LP1
                SEI
                RET
;=================
INT_0:CLI
                MOV R21,R17
                ANDI R21,1
                BRNE TRN
                MOV R19,R18
                ANDI R19,1
                BRNE MOD35H
                INC R22
                CPI R22,0X15
                BRLT LK1
                LDI R22,0
                RJMP LK1
MOD35H:         INC R22
                CPI R22,0X23
                BRNE LK1
                LDI R22,0
                RJMP LK1
TRN:   MOV R19,R18
                ANDI R19,1
                BRNE MD35H
                DEC R22
                CPI R22,0XFF
                BRNE LK1
                LDI R22,0X14
                RJMP LK1
MD35H:          DEC R22
                CPI R22,0XFF
                BRNE LK1
                LDI R22,0X22
LK1:   SEI
                RETI

INT_1:CLI
                INC R18
                SEI
                RETI

INT_2:CLI
                INC R17
                SEI
                RETI
;==================
.ORG 0X80
POLA:
```

```
.DB 0XC0,0XF9,0XA4,0XB0,0X99,0X92,0X82,0XF8
.DB 0X80,0X90,0X88,0X83,0XC6,0XA1,0X86,0X8E
```

